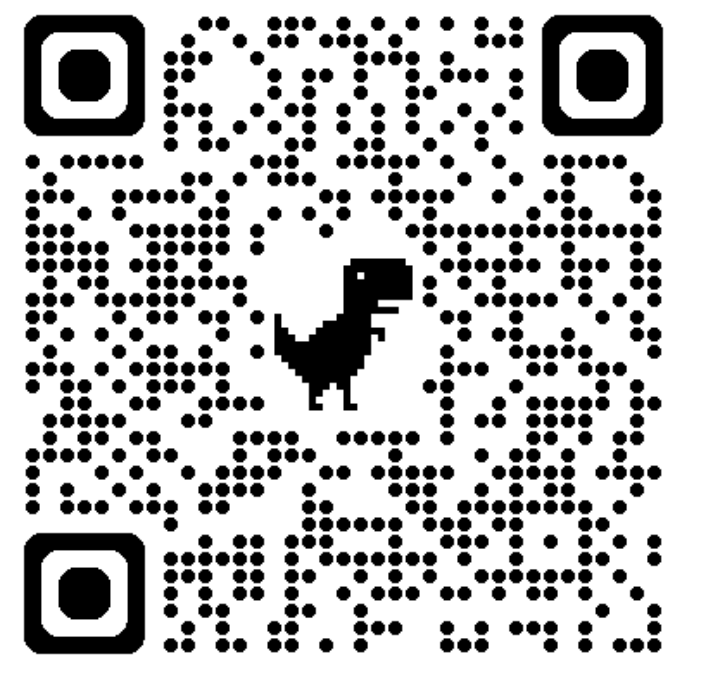


Multipartite entanglement characterization in quantum networks from spin clusters

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1. Introduction

- Large engineered quantum networks are difficult to characterize directly.
- Physical spin models provide tractable alternatives for characterization.
- Multipartite entanglement needed for protocols like CKA.
- Prior work [1] characterized only the pairwise layer.
- We study multipartite structure providing hyperdegree and hyperedge-size statistics here.

2. Model & Construction

- Random transverse-field Ising model.

$$\mathcal{H} = - \sum_{\langle ij \rangle} J_{ij} \sigma_i^x \sigma_j^x - \sum_i h_i \sigma_i^z \quad (1)$$

- $J_{ij} \in [0, 1]$, $h_i = h$; critical point $\theta_c = -0.17034$.
- SDRG [2] gives ground state as product of independent GHZ clusters:

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|\uparrow \cdots \uparrow\rangle \pm |\downarrow \cdots \downarrow\rangle). \quad (2)$$

- Circular nodes from $P(r) \propto r^{-2.67}$; spiral-packed [3] to 30% coverage.
- Hyperedge of size k = GHZ cluster overlapping 2 or more nodes
- Hyperdegree d = number of distinct GHZ clusters overlapping a node.

RTIM $\xrightarrow{\text{SDRG}}$ GHZ clusters $\longrightarrow$ Node packing $\xrightarrow{\text{Hyperedge}}$ Emergent network

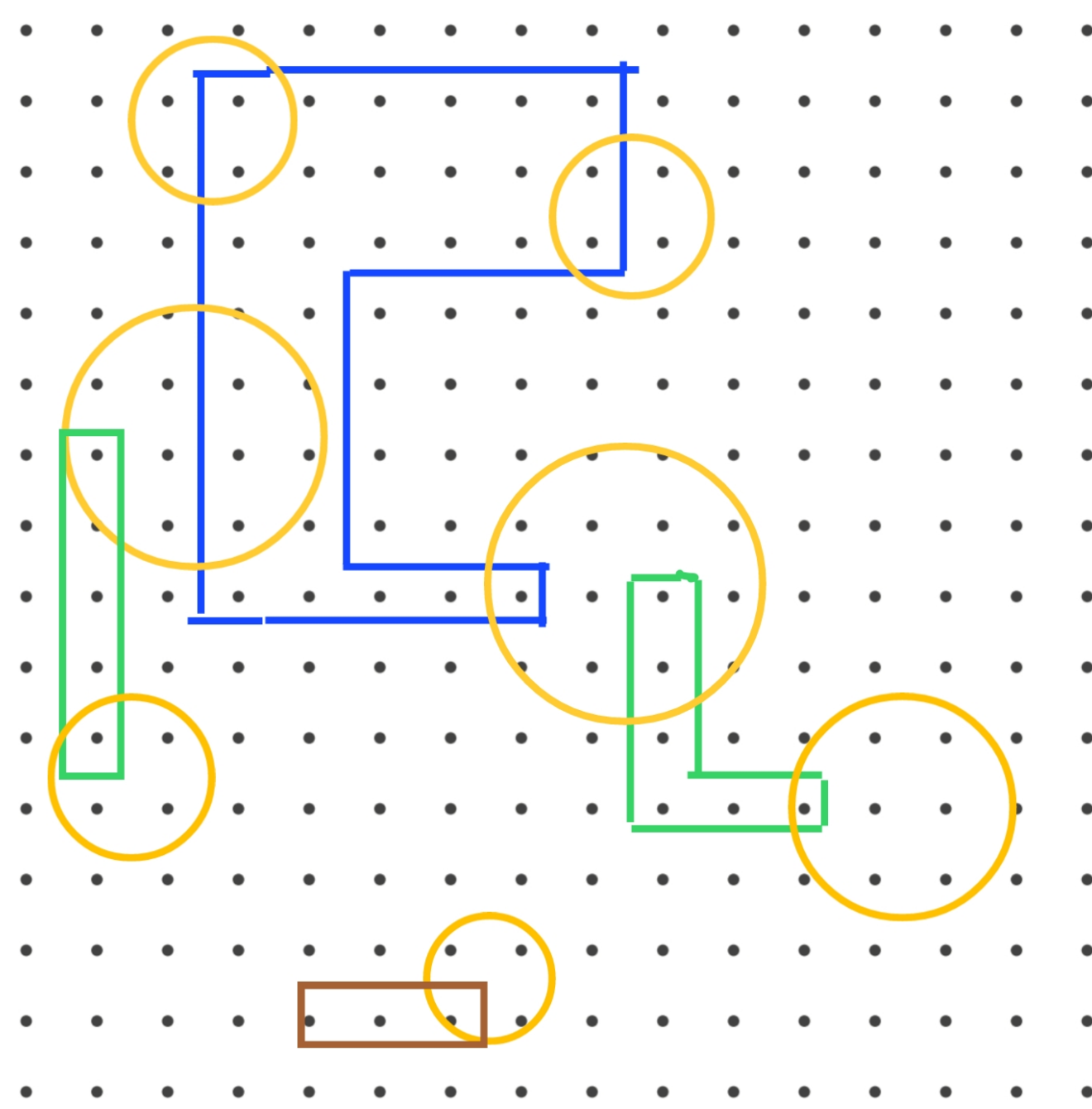


Fig. 1 Green = bipartite edge; blue = multipartite hyperedge; brown cluster touches only one node and contributes no hyperedge.

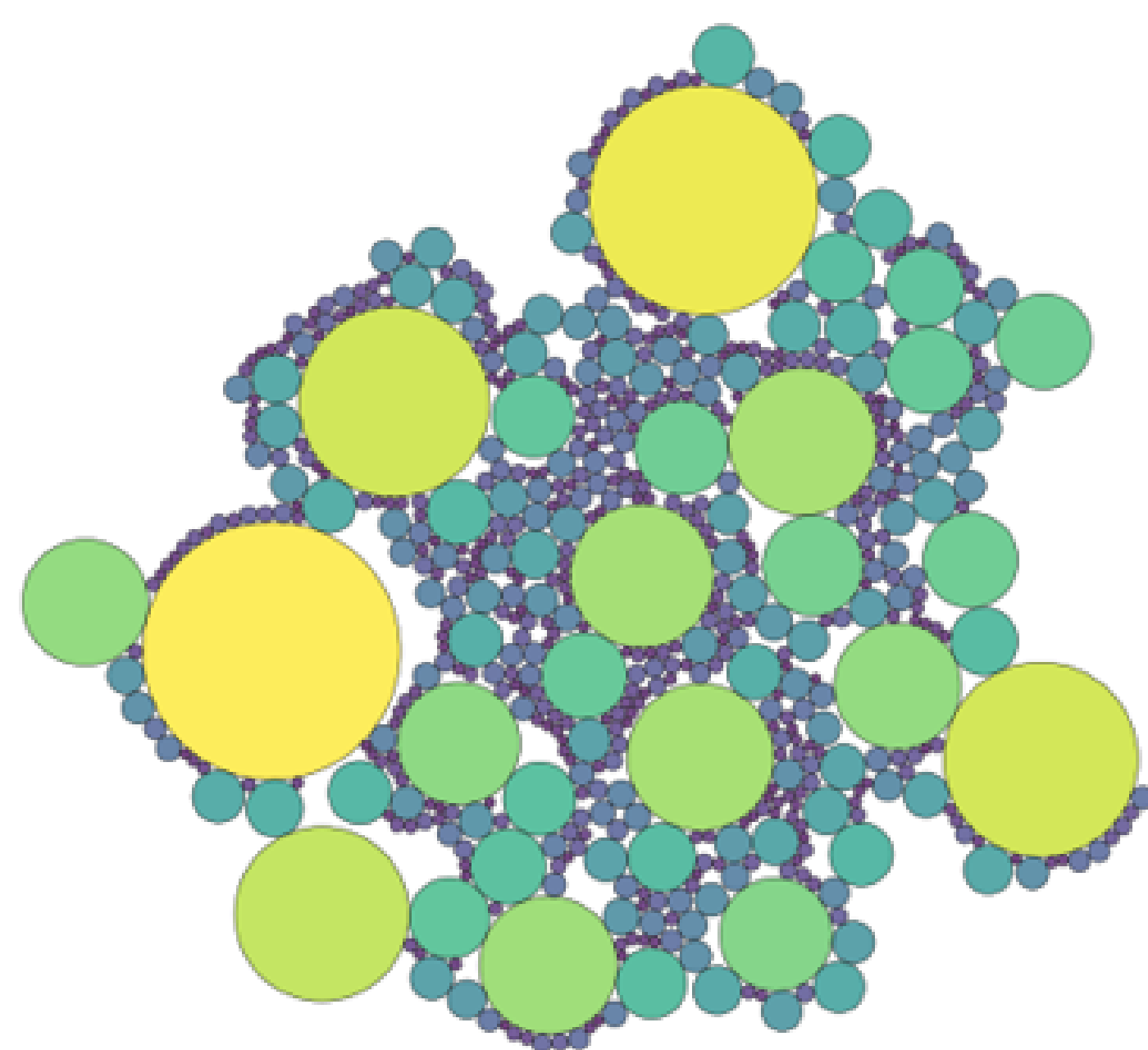


Fig. 2 Spiral-packed node geometry; node size reflects the assigned radius distribution $P(r) \propto r^{-2.67}$.

3. Results & Key Takeaways

- Size-independent fractions up to $L = 4096$:

~10%
of hyperedges are
multipartite

~40%
of nodes participate in
multipartite hyperedges

- **Hyperdegree distribution** is heavy-tailed and scale free ($2 < \gamma < 3$).
- **Hyperedge size distribution** is heavy-tailed.

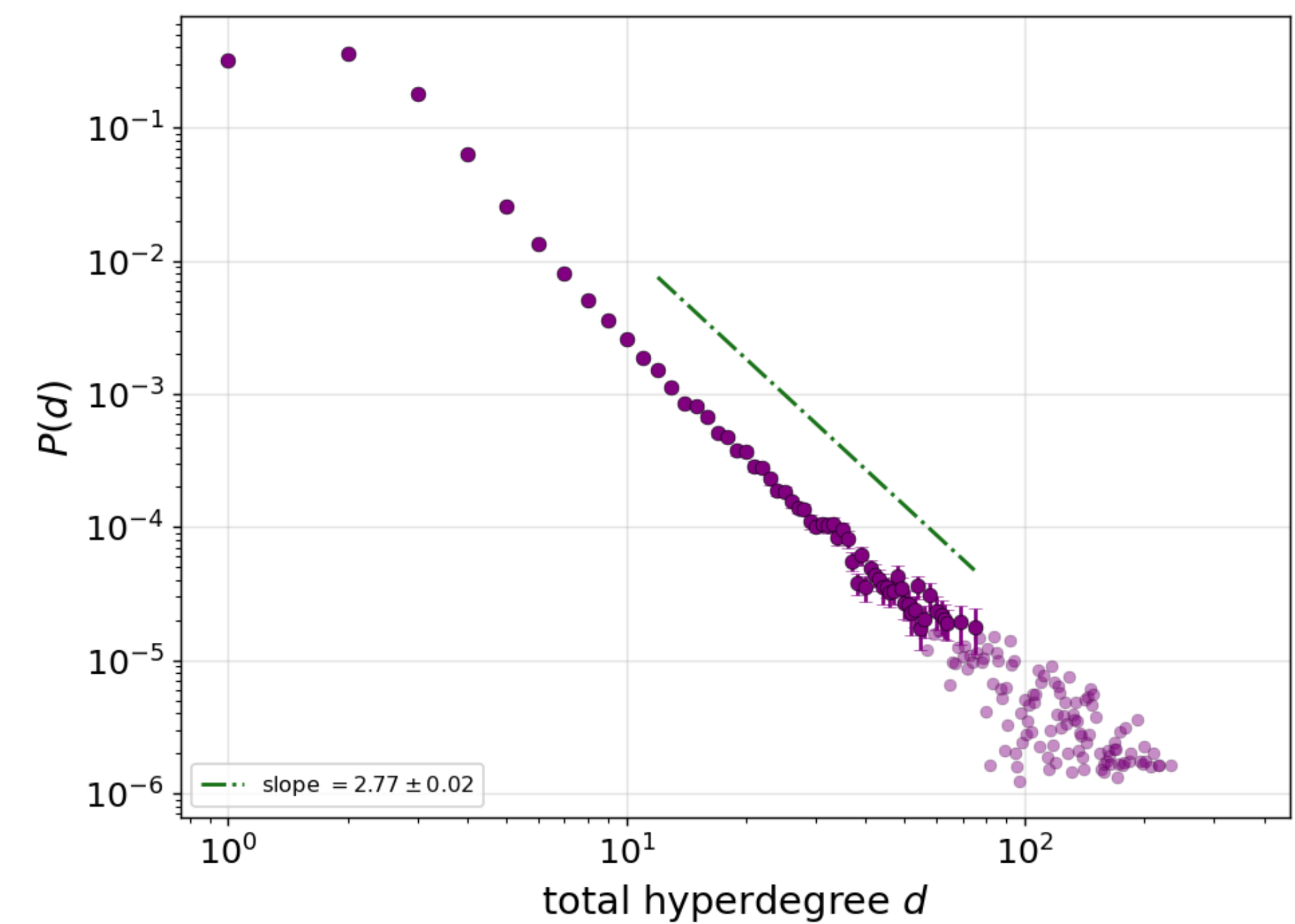


Fig. 3 Hyperdegree distribution

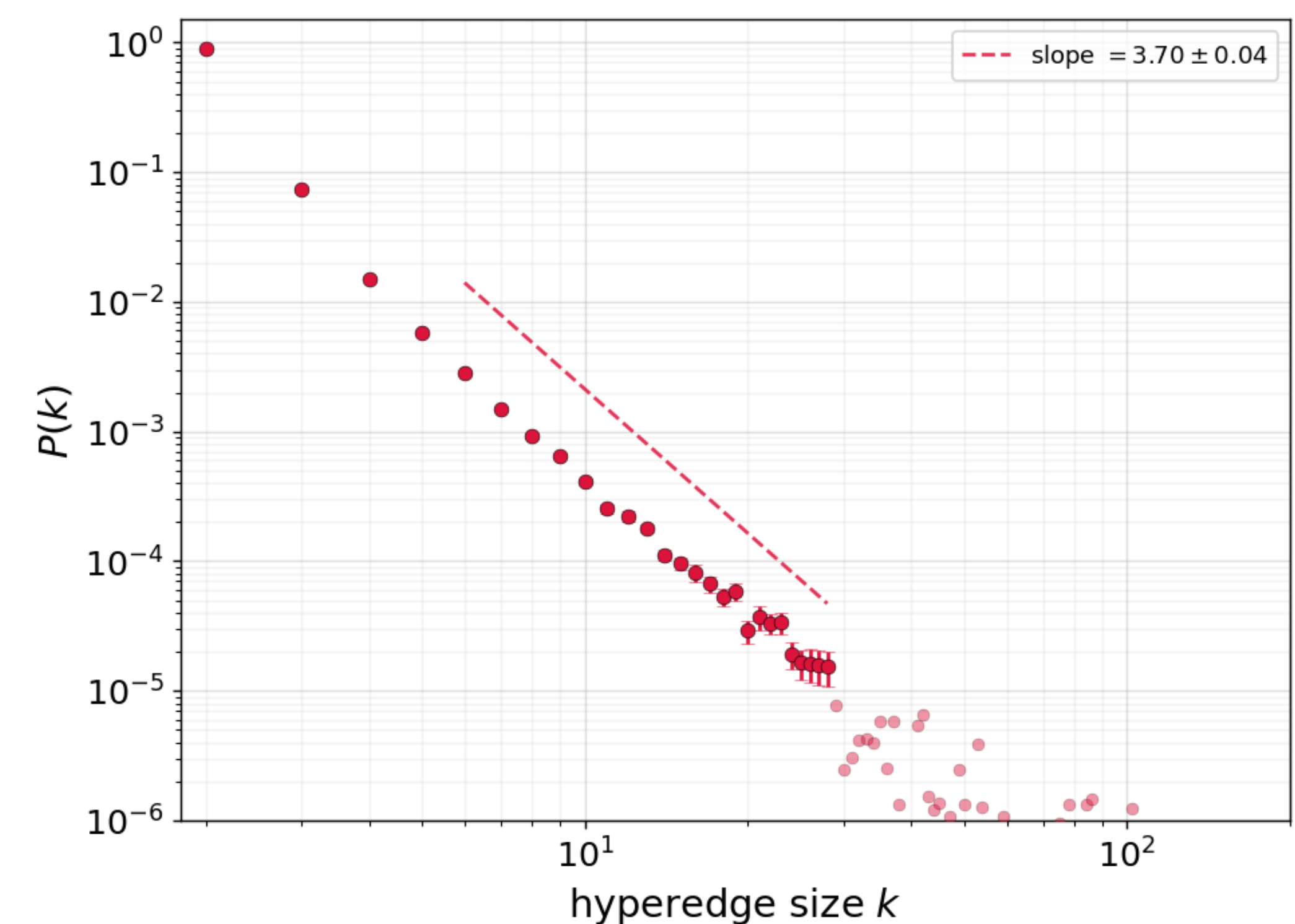


Fig. 4 Hyperedge-size distribution

- **Takeaways:** Simple spin models contain rich multipartite structure beyond the pairwise level.
- They could provide testbeds for study of the higher-order structure of quantum networks at tractable scales.
- **Future work:** Robustness and long-range connectivity of the multipartite network, and its usefulness for multiparty quantum communication protocols.

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